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Modeling the response of polymer–ionic liquid electromechanical actuators

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Abstract A model is derived for the electromechanical response of a porous membrane swollen with an ionic liquid and sandwiched between two nanoscale-thin electrodes under DC current. Bending of the membrane is induced by pressure in pores arising due to diffusion of ions through a network of nanochannels. Transport of ions is governed by the applied electric field and redox reactions at the surfaces of electrodes. Constitutive equations for the mechanical response of a porous medium and diffusion of ions are derived by means of the free energy imbalance inequality under an arbitrary deformation with finite strains. Under the assumption regarding small strains, but finite changes in concentrations of ions and the electrostatic potential, an explicit expression is developed for the curvature of the membrane. A steady-state solution to the Poisson–Nernst–Planck equations is obtained by means of the method of matched asymptotic expansions. Results of numerical analysis demonstrate the ability of the constitutive equations to describe observations. In particular, the model provides an explanation for bending to the anode and to the cathode and predicts qualitatively the effects of applied voltage, concentration of ionic liquid, and thickness of a membrane on its curvature.

1 Introduction

This study deals with constitutive modeling and numerical simulation of the elastic response of polymer–ionic liquid electromechanical actuators.

Functional polymer materials with strong electromechanical properties have become a focus of attention in the past decade due to their ability to serve as transducers (where mechanical stimulation generates a voltage) and actuators (where an applied voltage induces a mechanical response). Ionic polymer metal composites (IPMCs) are introduced as an important class of functional composite materials whose mechanical response (bending) is driven by transport of ions through a polymer electrolyte membrane sandwiched between two nanoscale-thin electrodes [1–5]. The basic element of IPMCs is a porous membrane manufactured of a polymer containing functional groups attached to main or side chains (Nafion provides the classical example of perfluorosulfonic acid ionomers, but several block polymers with a higher electroactuation ability and tailored microstructure were recently invented [6–10]). Before actuation, the membrane is soaked in an aqueous solution of an electrolyte, which leads to ionization of functional groups that dissociate into mobile cations and fixed anions. When a voltage is applied to the electrodes, cations move to the negative electrode inducing electroosmotic drag of neutral solvent molecules, while bound anions remain fixed. This leads to formation of a pressure gradient, which, in turn, results in noticeable deformation of the membrane at low voltages (1–4 V).

To avoid such drawbacks of IPMCs with liquid electrolytes as evaporation and leakage of solvent that lead to earlier dehydration of the membrane, aqueous electrolytes are replaced with ionic liquids [11, 12].

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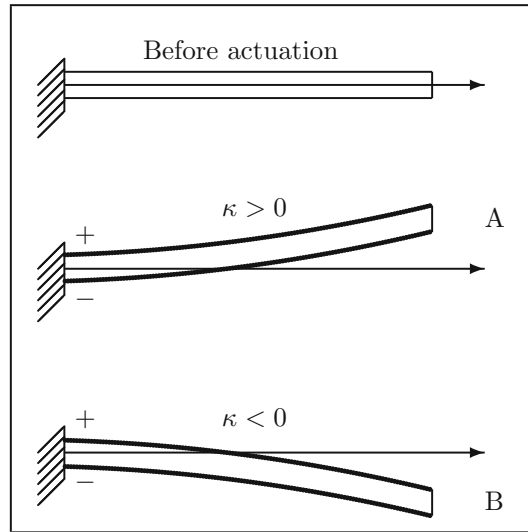


Fig. 1 Bending of an electromechanical actuator driven by an applied voltage

Ionic liquids (ILs) are fluid semi-organic salts composed of organic cations and organic or inorganic anions with melting points below 100 °C [13]. Due to their chemical and thermal stabilities, low vapor pressure, and high ionic conductivity, replacement of water with ILs induces a strong enhancement of electromechanical performance [7–10, 14–16].

Observations on electroactuation of porous membranes swollen with ILs show that immediately after application of DC current to the electrodes, the membrane bends to the anode, and its curvature κ grows with time t until it reaches some maximum value. Two cases are distinguished in the further development of curvature: (i) κ decreases slightly with time and approaches a positive steady-state value (case A in Fig. 1) and (ii) κ decays strongly with time (the back relaxation process [9]) and levels off at a negative value (case B in Fig. 1). The former type of response was reported in tests on copolymer membranes filled with ionic liquids [6, 7, 9, 10, 16, 17] and IPMCs with Flemion membranes [18], while the latter was observed in experiments on IPMCs with Nafion membranes swollen with various solvents [19], and Nafion and Aquivion membranes soaked in ionic liquids [10, 17, 20]. Although physical mechanisms governing the time-dependent behavior of actuators swollen with ILs are still under debate, it is generally accepted that their performance is strongly affected by electrochemistry of interactions between mobile ions and the surfaces of electrodes [21, 22]. This conclusion is confirmed by experimental data [7, 23], which show that replacement of one type of cations in an IL with another changes the sign of curvature under steady-state conditions, whereas chemical modification of anions results in modest quantitative changes in steady-state curvature only [24].

Experimental investigation of the effects of external parameters (voltage V , concentration of ionic liquid C_0 , and thickness of a membrane H_0) on the steady-state curvature κ_s leads to the following conclusions:

1. Curvature of an actuator κ_s increases linearly with voltage V at very small applied potentials and approaches its ultimate value at relatively large voltages. In between these extremes, the function $\kappa_s(V)$ demonstrates either a sub-linear growth up to saturation [6, 7, 14, 17, 25] or an S -shaped behavior with an interval of super-linear increase [26–28].
2. At reasonably small concentrations of ionic liquid in a porous membrane C_0 (<20–40 wt%), κ_s grows linearly with C_0 . This increase is enhanced by applied voltage V [25], is independent of direction of bending [29], and is accompanied by an increase in the rate of bending [30]. At higher degrees of swelling of a membrane with an ionic liquid, a strong decay in κ_s with C_0 is reported [31, 32].
3. The steady-state curvature increases with thickness H_0 of a porous membrane. In [33], a linear growth of κ_s with H_0 is reported when a fixed voltage is applied to electrodes. A nonlinear increase in κ_s with thickness is demonstrated in [34] where it is shown that this growth is strongly enhanced by applied potentials. To the best of our knowledge, the observed effect has not yet been explained in detail. An increase in curvature with thickness of a membrane contradicts experimental data in conventional bending tests (when external forces are applied to an elastic membrane, deformation is reduced with an increase in stiffness), as well as to predictions of standard models for bending of a membrane in an electric field [35].

The objective of this study is to develop a constitutive model for the electromechanical response of an actuator able to describe these phenomena.

The first model for electroactuation of a porous membrane swollen with an aqueous solution of a poly-electrolyte was suggested in [36]. Bending of a membrane is driven by transport of two species, neutral water molecules and positive ions, whose fluxes are connected with the electric field vector and the gradient of pressure by linear equations (the generalized Onsager relations). When a DC current is applied, diffusion of positive ions to the negative electrode results in the development of solvent pressure, which, in turn, leads to deformation of the membrane. With reference to this approach, analysis of bending of polyelectrolyte gels and ionogels under the action of an electric field was performed in [37,38].

A constitutive framework for the analysis of electroactuation of IPMCs has been developed in [39–50], to mention but a few. Most of these studies focus on the time-dependent response at the initial interval of deformation (immediately after application of voltage), presume diffusivity of positive ions to exceed strongly that of negative ions, and explain bending of a membrane by the action of ionic pressure developed in a domain rich in positive ions near the negative electrode. Although observations confirm that diffusivity of cations in a nanoporous medium is substantially higher than diffusivity of anions [51], their ratio depends strongly on concentration of ions, structure of nanochannels between pores, and distribution of bound charges at the surfaces of pores, which requires evolution of a network of nanochannels to be accounted explicitly [17,52,53].

It has recently been accepted that interactions between mobile ions and surfaces of the electrodes play the key role in the electromechanical response of IPMCs. Four types of phenomena are distinguished: (i) nanoscale morphology of electrodes prepared by electroless plating process [22,54–56], (ii) steric interactions between finite-sized ions crowding in the vicinity (diffuse part of the boundary layer) of electrodes [57,58], (iii) depletion in ion concentration next to (compact part of the boundary layer) the interface between a porous membrane and electrodes (the so-called concentration polarization) [59], and (iv) redox reactions between mobile ions and metal atoms at the surfaces of electrodes [60–62].

In the present study, an electromechanical actuator is modeled as a porous compressible membrane swollen with an ionic liquid, which is presumed to be dissociated entirely into ions (formation of ion pairs by positive and negative ions is disregarded). Bending rigidity of nanoscale-thin electrodes is neglected (this parameter can be accounted for by recalculation of coefficients in the expression for curvature). Under application of electric potentials to the electrodes, transport of ions starts through a network of nanochannels connecting pores. A model with two transport mechanisms (through nanochannels and the bulk medium) was recently proposed in [63]. Diffusion of ions induces changes in ionic pressure in pores that causes deformation of the membrane, which, in turn, affects its porosity. Volume deformation of a porous medium is connected with its porosity by a simple differential equation. More sophisticated relations applicable for the analysis of bending of porous membranes filled with liquids were recently developed in [64,65]. Distribution of negative ions in the membrane obeys the mass conservation law, whereas concentration of positive ions is governed by the Butler–Volmer equations for redox reactions at the electrodes. To simplify the analysis, we presume cations to be converted into neutral atoms immobilized at the surfaces of electrodes. Diffusion of neutral species produced due to the redox reactions was accounted for in [66]. An explicit solution to the steady-state Poisson–Nernst–Planck problem for distribution of ions and the electrostatic potential is derived by means of the asymptotic expansion technique [67]. This method has recently been applied for the analysis of bending of IPMCs in [47,49,54,68].

The exposition is organized as follows: Constitutive equations for the elastic response of a porous medium induced by diffusion of ions are developed in Sect. 2. After simplification for the case of small strains, these relations are applied in Sect. 3 to study bending of a porous membrane. To derive an explicit expression for its curvature under steady-state deformation, Poisson–Nernst–Planck equations are solved in the “Appendix”. The effect of an electric field on bending of a membrane is analyzed for small and large ratios of thicknesses of the Stern (compact) and diffuse layers. Comparison of observations with results of simulation is performed in Sect. 4. Concluding remarks are formulated in Sect. 5.

2 Constitutive model

A membrane soaked with an ionic liquid is modeled as a two-phase continuum consisting of a solid phase (an isotropic porous compressible medium) and mobile ions located in a network of pores. Transport of ions is treated as their diffusion through a system of nanochannels accelerated by an electric field.

2.1 Macro-deformation of a porous medium

The initial state of a porous medium coincides with its undeformed state. Transformation of the initial configuration into the actual configuration is described by the deformation gradient $\mathbf{F}$. The Cauchy–Green tensors for macro-deformation read

$$\mathbf{B} = \mathbf{F} \cdot \mathbf{F}^\top, \quad \mathbf{C} = \mathbf{F}^\top \cdot \mathbf{F}, \quad (1)$$

where the dot stands for inner product, and $\top$ denotes transpose.

The volume element in the actual configuration dv is expressed by means of the volume element in the initial configuration dV as

$$dv = J dV, \quad J = \det \mathbf{F}. \quad (2)$$

The surface element $\mathbf{n} da$ with unit normal $\mathbf{n}$ in the actual configuration is connected with the surface element $\mathbf{N} dA$ with unit normal $\mathbf{N}$ in the initial configuration by the equation

$$\mathbf{n} da = J \mathbf{F}^{-\top} \cdot \mathbf{N} dA = J \mathbf{N} \cdot \mathbf{F}^{-1} dA. \quad (3)$$

Porosity of the medium is determined as

$$\gamma = \frac{dv_p}{dv} \quad (4)$$

where dv_p denotes the volume occupied by pores in the elementary volume dv .

Denote by C^+ , C^- concentrations of positive and negative ions (numbers of species per unit volume) in the initial configuration. Their concentrations per unit volume in the actual configuration are given by $c^+ = C^+/J$, $c^- = C^-/J$. As ions are located in pores exclusively, their densities in the actual state read

$$\rho^+ = \frac{C^+}{\gamma J}, \quad \rho^- = \frac{C^-}{\gamma J}. \quad (5)$$

2.2 Electric field

The electrostatic potential Φ is connected with the electric field vectors in the actual, $\mathbf{e}$, and initial, $\mathbf{E}$, states by the equations

$$\mathbf{e} = -\nabla \Phi, \quad \mathbf{E} = -\nabla_0 \Phi \quad (6)$$

where ∇ and ∇_0 are the gradient operators in the actual and initial configurations. A porous medium is modeled as a linear dielectric material whose electric displacement vector in the actual state reads

$$\mathbf{h} = \varepsilon \mathbf{e}, \quad (7)$$

where the constant ε stands for the electric permittivity. It follows from Eqs. (6), (7) and the chain rule for differentiation,

$$\nabla_0 \Phi = \nabla \Phi \cdot \mathbf{F} = \mathbf{F}^\top \cdot \nabla \Phi, \quad (8)$$

that

$$\mathbf{E} = \mathbf{e} \cdot \mathbf{F} = \mathbf{F}^\top \cdot \mathbf{e}, \quad \mathbf{h} = \varepsilon \mathbf{E} \cdot \mathbf{F}^{-1} = \varepsilon \mathbf{F}^{-\top} \cdot \mathbf{E}. \quad (9)$$

The nominal electric displacement reads

$$\mathbf{H} = J \mathbf{F}^{-1} \cdot \mathbf{h} = J \mathbf{h} \cdot \mathbf{F}^{-\top}. \quad (10)$$

It follows from Eqs. (1), (9), (10) that

$$\mathbf{H} = \varepsilon J \mathbf{C}^{-1} \cdot \mathbf{E} = \varepsilon J \mathbf{E} \cdot \mathbf{C}^{-1}. \quad (11)$$

The Gauss law for a dielectric medium implies that vector $\mathbf{H}$ obeys the equation

$$\nabla_0 \cdot \mathbf{H} = R \quad (12)$$

where $R = e(C^+ - C^-)$ denotes charge density in the initial state, and $e = 1.602 \cdot 10^{-19}$ C stands for the charge of electron. Combination of Eqs. (6), (11) (12) results in the Poisson equation

$$\nabla_0 \cdot (\varepsilon J C^{-1} \cdot \nabla_0 \Phi) = -e(C^+ - C^-). \quad (13)$$

For an ideal dielectric medium, the energy density of the electric field (per unit volume in the initial configuration) reads [44]

$$W_{\text{el}} = \frac{1}{2\varepsilon J} \mathbf{H} \cdot \mathbf{C} \cdot \mathbf{H}. \quad (14)$$

For an arbitrary domain Ω with boundary $\partial\Omega$, the work of the electric field (per unit time) is determined by [44]

$$\mathcal{U}_{\text{el}} = - \int_{\partial\Omega} \Phi \dot{\mathbf{H}} \cdot \mathbf{N} dA$$

where the superscript dot stands for the derivative with respect to time. Transforming the integral by means of the Stokes formula and applying Eqs. (6), (12), we find that the work of electric field (per unit volume in the initial configuration and unit time) is given by

$$u_{\text{el}} = \mathbf{E} \cdot \dot{\mathbf{H}} - e\Phi(\dot{C}^+ - \dot{C}^-). \quad (15)$$

2.3 Kinetic relations

Denote by $\mathbf{J}^+$, $\mathbf{J}^-$ and $\mathbf{j}^+$, $\mathbf{j}^-$ the flux vectors for mobile ions (numbers of ions moving through unit area per unit time) in the initial and actual configurations. Keeping in mind that $\mathbf{n} \cdot \mathbf{j} da = \mathbf{N} \cdot \mathbf{J} dA$, we find from Eq. (3) that

$$\mathbf{J}^+ = J \mathbf{F}^{-1} \cdot \mathbf{j}^+, \quad \mathbf{J}^- = J \mathbf{F}^{-1} \cdot \mathbf{j}^-. \quad (16)$$

The flux vectors in the actual configuration are connected with chemical potential of mobile ions, μ^+ and μ^- , by the formulas

$$\mathbf{j}^+ = -\frac{D^+ c^+}{k_B T} \nabla \mu^+, \quad \mathbf{j}^- = -\frac{D^- c^-}{k_B T} \nabla \mu^- \quad (17)$$

where T is the absolute temperature, k_B is Boltzmann's constant, and D^+ , D^- denote diffusivities. Presuming for simplicity that

$$D^+ = D^- = D,$$

where D is a constant, we find from Eqs. (8), (16), (17) that

$$\mathbf{J}^+ = -\frac{DC^+}{k_B T} \mathbf{F}^{-1} \cdot \nabla_0 \mu^+ \cdot \mathbf{F}^{-1}, \quad \mathbf{J}^- = -\frac{DC^-}{k_B T} \mathbf{F}^{-1} \cdot \nabla_0 \mu^- \cdot \mathbf{F}^{-1}. \quad (18)$$

Insertion of Eq. (18) into the mass conservation laws for ions,

$$\dot{C}^+ = -\nabla_0 \cdot \mathbf{J}^+, \quad \dot{C}^- = -\nabla_0 \cdot \mathbf{J}^-, \quad (19)$$

results in the diffusion equations

$$\dot{C}^+ = \nabla_0 \cdot \left(\frac{DC^+}{k_B T} \mathbf{F}^{-1} \cdot \nabla_0 \mu^+ \cdot \mathbf{F}^{-1} \right), \quad \dot{C}^- = \nabla_0 \cdot \left(\frac{DC^-}{k_B T} \mathbf{F}^{-1} \cdot \nabla_0 \mu^- \cdot \mathbf{F}^{-1} \right). \quad (20)$$

For an arbitrary domain Ω with boundary $\partial\Omega$, the work produced by transport of ions per unit time reads [44]

$$\mathcal{U}_{\text{dif}} = - \int_{\partial\Omega} (\mu^+ \mathbf{J}^+ + \mu^- \mathbf{J}^-) \cdot \mathbf{N} dA.$$

Transforming the integral by means of the Stokes formula and using Eqs. (18), (19), we find that the work produced by diffusion of ions (per unit volume in the initial configuration and unit time) is given by

$$u_{\text{dif}} = \mu^+ \dot{C}^+ + \mu^- \dot{C}^- + \bar{u}_{\text{dif}} \quad (21)$$

where

$$\bar{u}_{\text{dif}} = \frac{D}{k_B T} \left[C^+ (\mathbf{F}^{-1} \cdot \nabla_0 \mu^+) \cdot (\mathbf{F}^{-1} \cdot \nabla_0 \mu^+) + C^- (\mathbf{F}^{-1} \cdot \nabla_0 \mu^-) \cdot (\mathbf{F}^{-1} \cdot \nabla_0 \mu^-) \right] \geq 0. \quad (22)$$

2.4 Kinematic relations

Denote by

$$\mathbf{L} = \dot{\mathbf{F}} \cdot \mathbf{F}^{-1}, \quad \mathbf{D} = \frac{1}{2}(\mathbf{L} + \mathbf{L}^\top) \quad (23)$$

the velocity gradient and the rate-of-strain tensor for macro-deformation. The derivatives of the principal invariants J_1, J_2, J_3 of the Cauchy–Green tensors (1) read

$$\dot{J}_1 = 2\mathbf{B} : \mathbf{D}, \quad \dot{J}_2 = 2(J_2 \mathbf{I} - J_3 \mathbf{B}^{-1}) : \mathbf{D}, \quad \dot{J}_3 = 2J_3 \mathbf{I} : \mathbf{D}, \quad (24)$$

where the colon stands for convolution of tensors.

To describe the evolution of porosity induced by macro-deformation, we presume γ and J to be connected by the equation

$$\dot{\gamma} = a \frac{\dot{J}}{J} \quad (25)$$

where a is a constant coefficient. It will be shown in what follows that Eq. (25) implies conventional formula (43) for ionic pressure, where a is responsible for compressibility of the ionic gas.

2.5 Free energy density

The Helmholtz free energy Ψ of a two-phase continuum (per unit volume in the initial configuration) is treated as the sum of four components: (i) the energy of ions not interacting with each other and with the solid phase Ψ_1 , (ii) the energy of the solid phase not interacting with ions Ψ_2 , (iii) the energy of mixing of ions in pores Ψ_3 , and (iv) the energy of the electric field W_{el} ,

$$\Psi = \Psi_1 + \Psi_2 + \Psi_3 + W_{\text{el}}. \quad (26)$$

The specific energy density of ions is given by

$$\Psi_1 = \mu_0^+ C^+ + \mu_0^- C^- \quad (27)$$

where μ_0^+, μ_0^- denote chemical potentials of non-interacting ions.

The free energy density of an isotropic porous medium reads

$$\Psi_2 = W(J_1, J_2, J_3) \quad (28)$$

where W is a given function of the principal invariants of the Cauchy–Green tensor.

In the approximation of a dilute solution, the specific energy of mixing of ions in pores is determined by

$$\Psi_3 = k_B T \left[C^+ \left(\ln \frac{C^+}{C_*} - 1 \right) + C^- \left(\ln \frac{C^-}{C_*} - 1 \right) \right] \quad (29)$$

where C_* stands for the number of sites (per unit volume in the initial configuration) available for ions. This quantity is expressed in terms of porosity γ as

$$C_* = \frac{\gamma}{v} \quad (30)$$

where v is the characteristic volume of an ion.

2.6 Free energy imbalance inequality

Under isothermal deformation with finite strains, the free energy imbalance inequality reads

$$\dot{\Psi} - u_{\text{mech}} - u_{\text{el}} - u_{\text{dif}} \leq 0 \quad (31)$$

where $u_{\text{mech}}, u_{\text{el}}, u_{\text{dif}}$ are works (per unit volume in the initial state and unit time) produced by stresses, electric field, and transport of ions.

The mechanical work is determined by the conventional formula

$$u_{\text{mech}} = J \mathbf{T} : \mathbf{D} \quad (32)$$

where $\mathbf{T}$ is the Cauchy stress tensor. The works performed by the electric field and diffusion of cations are given by Eqs. (15) and (21), respectively.

It follows from Eq. (24) that the derivative of Eq. (28) with respect to time reads

$$\dot{W} = 2\mathbf{K} : \mathbf{D} \quad (33)$$

where

$$\mathbf{K} = W_{,1}\mathbf{B}_e - J_{e3}W_{,2}\mathbf{B}_e^{-1} + (J_{e2}W_{,2} + J_{e3}W_{,3})\mathbf{I}, \quad W_{,m} = \frac{\partial W}{\partial J_m}. \quad (34)$$

Differentiating Eq. (14) with respect to time and using Eq. (11), we find that

$$\dot{W}_{\text{el}} = \mathbf{E} \cdot \dot{\mathbf{H}} + \frac{1}{2\varepsilon J} \left(\mathbf{H} \cdot \dot{\mathbf{C}} \cdot \mathbf{H} - \frac{j}{J} \mathbf{H} \cdot \mathbf{C} \cdot \mathbf{H} \right).$$

Keeping in mind that

$$\dot{\mathbf{C}} = 2\mathbf{F}^\top \cdot \mathbf{D} \cdot \mathbf{F}, \quad j = J\mathbf{I} : \mathbf{D} \quad (35)$$

and using Eqs. (10), (11), we obtain

$$\dot{W}_{\text{el}} = \mathbf{E} \cdot \dot{\mathbf{H}} + J\mathbf{T}_M : \mathbf{D} \quad (36)$$

where

$$\mathbf{T}_M = \frac{1}{\varepsilon} \left[(\mathbf{h} \otimes \mathbf{h}) - \frac{1}{2} (\mathbf{h} \cdot \mathbf{h}) \mathbf{I} \right] \quad (37)$$

denotes the Maxwell stress [69], and $\otimes$ stands for the tensor product.

It follows from Eqs. (25), (27), (29), (30), (35) that

$$\dot{\Psi}_1 = \mu_0^+ \dot{C}^+ + \mu_0^- \dot{C}^-, \quad \dot{\Psi}_3 = k_B T \left[\dot{C}^+ \ln \frac{C^+ v}{\gamma} + \dot{C}^- \ln \frac{C^- v}{\gamma} - a \frac{C^+ + C^-}{\gamma} (\mathbf{I} : \mathbf{D}) \right]. \quad (38)$$

Substitution of Eqs. (15), (21), (26), (32), (33), (36), (38) into Eq. (31) implies that

$$\begin{aligned} & \left(2\mathbf{K} + J\mathbf{T}_M - ak_B T \frac{C^+ + C^-}{\gamma} \mathbf{I} - J\mathbf{T} \right) : \mathbf{D} + \left(\mu_0^+ + k_B T \ln \frac{C^+ v}{\gamma} + e\Phi - \mu^+ \right) \dot{C}^+ \\ & + \left(\mu_0^- + k_B T \ln \frac{C^- v}{\gamma} - e\Phi - \mu^- \right) \dot{C}^- - \bar{u}_{\text{dif}} \leq 0. \end{aligned} \quad (39)$$

Bearing in mind that $\mathbf{D}$, C^+ , C^- are arbitrary functions and utilizing Eq. (34), we conclude that Eq. (39) is fulfilled, provided that (i) the Cauchy stress tensor reads

$$\mathbf{T} = \frac{2}{J} \left[W_{,1} \mathbf{B} - J_3 W_{,2} \mathbf{B}^{-1} + (J_2 W_{,2} + J_3 W_{,3}) \mathbf{I} \right] + \mathbf{T}_M - P \mathbf{I}, \quad (40)$$

and (ii) chemical potentials of ions are given by

$$\mu^+ = \mu_0^+ + k_B T \ln \frac{C^+ v}{\gamma} + e\Phi, \quad \mu^- = \mu_0^- + k_B T \ln \frac{C^- v}{\gamma} - e\Phi \quad (41)$$

where

$$P = ak_B T \frac{C^+ + C^-}{\gamma J}. \quad (42)$$

Combination of Eqs. (5) and (42) implies the formula for ionic pressure P of an ideal gas of non-interacting species

$$P = ak_B T (\rho^+ + \rho^-). \quad (43)$$

Constitutive equations for an isotropic porous medium under finite deformation involve stress–strain relations (37), (40), (42) and formulas (41) for chemical potentials of ions. These relations are to be accompanied by (i) the equilibrium equation for the Cauchy stress tensor, (ii) kinetic Eq. (25) for porosity, (iii) diffusion equations (20), and (iv) Eq. (13) for the electrostatic potential, as well as appropriate initial and boundary conditions.

2.7 Governing equations at small strains

We now focus on the case of small strains and finite changes in concentrations of ions and electrostatic potential in a porous medium. Integration of Eq. (25) with the initial conditions $\gamma(0) = \gamma_0$, $J(0) = 1$ implies that

$$\gamma = \gamma_0 + a \ln J. \quad (44)$$

Keeping in mind that at small strains

$$J = 1 + (\mathbf{I} : \mathcal{E}) \quad (45)$$

where $\mathcal{E}$ denotes the strain tensor, we find from Eq. (44) that

$$\gamma = \gamma_0 + a(\mathbf{I} : \mathcal{E}). \quad (46)$$

Disregarding small changes in porosity, we find from Eq. (41) that

$$\mu^+ = \mu_0^+ + k_B T \ln \frac{C^+}{C_{*0}} + e\Phi, \quad \mu^- = \mu_0^- + k_B T \ln \frac{C^-}{C_{*0}} - e\Phi \quad (47)$$

where

$$C_{*0} = \frac{\gamma_0}{v}.$$

To calculate the flux vectors, we insert Eq. (47) into Eq. (18) and neglect small terms,

$$\mathbf{J}^+ = -D \left(\nabla_0 C^+ + \frac{eC^+}{k_B T} \nabla_0 \Phi \right), \quad \mathbf{J}^- = -D \left(\nabla_0 C^- - \frac{eC^-}{k_B T} \nabla_0 \Phi \right). \quad (48)$$

It follows from Eq. (13) that, with the required level of accuracy, Φ is governed by the equation

$$\Delta_0 \Phi = -\frac{e}{\varepsilon} (C^+ - C^-) \quad (49)$$

where $\Delta_0 = \nabla_0 \cdot \nabla_0$ stands for the Laplace operator in the initial configuration. Combination of Eqs. (6), (9), (37) implies that

$$\mathbf{T}_M = \varepsilon \left[(\nabla_0 \Phi \otimes \nabla_0 \Phi) - \frac{1}{2} (\nabla_0 \Phi \cdot \nabla_0 \Phi) \mathbf{I} \right]. \quad (50)$$

According to Eqs. (42), (45), (46), we find that

$$P = ak_B T \frac{C^+ + C^-}{\gamma_0}. \quad (51)$$

The stress–strain relation (40) reads

$$\mathbf{T} = \mathbf{T}_E + \mathbf{T}_M - P \mathbf{I} \quad (52)$$

where

$$\mathbf{T}_E = 3K(\mathbf{I} : \boldsymbol{\mathcal{E}})\mathbf{I} + 2G\left(\boldsymbol{\mathcal{E}} - \frac{1}{3}(\mathbf{I} : \boldsymbol{\mathcal{E}})\mathbf{I}\right) \quad (53)$$

stands for the elastic stress in an isotropic medium, and K and G are bulk and shear moduli. To transform Eqs. (52) and (53), we split the tensor $\boldsymbol{\mathcal{E}}$ into the sum

$$\boldsymbol{\mathcal{E}} = \boldsymbol{\mathcal{E}}_0 + \boldsymbol{\epsilon} \quad (54)$$

where the strain tensor $\boldsymbol{\mathcal{E}}_0$ describes equilibrium swelling of a porous medium immersed into a bath with a ionic liquid, and $\boldsymbol{\epsilon}$ characterizes its additional deformation driven by diffusion of ions under application of an electric field.

Under homogeneous swelling of an isotropic medium, concentrations C^+ , C^- are constant and equal to C_0 , the Maxwell stress $\mathbf{T}_M$ equals zero, and the tensor $\boldsymbol{\mathcal{E}}_0$ is determined from the condition that the stress tensor $\mathbf{T}$ vanishes. It follows from Eqs. (51)–(54) that

$$\boldsymbol{\mathcal{E}}_0 = \frac{2ak_B T C_0}{3K\gamma_0} \mathbf{I}. \quad (55)$$

Insertion of Eqs. (51), (53)–(55) into Eq. (52) yields

$$\mathbf{T} = \frac{E\nu}{(1-2\nu)(1+\nu)}(\epsilon - p)\mathbf{I} + \frac{E}{1+\nu}\boldsymbol{\epsilon} + \mathbf{T}_M \quad (56)$$

where ϵ , $\mathbf{e}$ stand for the first invariant and deviator of the strain tensor $\boldsymbol{\epsilon}$,

$$\boldsymbol{\epsilon} = \frac{1}{3}\epsilon\mathbf{I} + \mathbf{e},$$

Young's modulus E and Poisson's ratio ν are given by

$$K = \frac{E}{3(1-2\nu)}, \quad G = \frac{E}{2(1+\nu)},$$

and the dimensionless pressure p reads

$$p = \frac{3ak_B T(1-2\nu)}{E\gamma_0}(C^+ + C^- - 2C_0). \quad (57)$$

3 Bending of a membrane

A porous membrane is modeled as a rectangular block that occupies the domain $\{-H \leq X_1 \leq H, -L \leq X_2 \leq L, -L_* \leq X_3 \leq L_*\}$ in the initial configuration. Here, X_m are Cartesian coordinates with unit vectors $\mathbf{i}_m$, and $2H$, $2L$, and $2L_*$ denote thickness, length, and width of the undeformed membrane, respectively. The membrane is swollen homogeneously in a bath with a ionic liquid until concentration of ions reaches a fixed value C_0 . At instant $t = 0$, electric potentials $-\frac{V}{2}$ and $\frac{V}{2}$ are applied to the electrodes at the boundaries $X_1 = -H$ and $X_1 = H$. The electric field induces transport of ions, which, in turn, leads to deformation of the membrane. Under the condition $L \gg L_* \gg H$, this deformation is treated as bending in the plane $\{X_1, X_2\}$. For simplicity, the electrodes are presumed to be very thin, and the effect of their bending rigidity on deformation of the membrane is disregarded.

3.1 Governing equations

We now focus on the one-dimensional problem and presume concentrations of ions C^+ , C^- and electrostatic potential Φ to depend on time t and coordinate X_1 only. It follows from Eq. (48) that $\mathbf{J}^+ = J^+ \mathbf{i}_1$, $\mathbf{J}^- = J^- \mathbf{i}_1$, where the quantities

$$J^+ = -D \left(\frac{\partial C^+}{\partial X_1} + \frac{eC^+}{k_B T} \frac{\partial \Phi}{\partial X_1} \right), \quad J^- = -D \left(\frac{\partial C^-}{\partial X_1} - \frac{eC^-}{k_B T} \frac{\partial \Phi}{\partial X_1} \right) \quad (58)$$

obey Eq. (19),

$$\dot{C}^+ = -\frac{\partial J^+}{\partial X_1}, \quad \dot{C}^- = -\frac{\partial J^-}{\partial X_1}. \quad (59)$$

It follows from Eqs. (49), (50) that the function Φ is governed by the equation

$$\frac{\partial^2 \Phi}{\partial X_1^2} = -\frac{e}{\varepsilon} (C^+ - C^-), \quad (60)$$

and the Maxwell stress reads

$$\mathbf{T}_M = \frac{\varepsilon}{2} \left(\frac{\partial \Phi}{\partial X_1} \right)^2 (\mathbf{i}_1 \otimes \mathbf{i}_1 - \mathbf{i}_2 \otimes \mathbf{i}_2 - \mathbf{i}_3 \otimes \mathbf{i}_3). \quad (61)$$

Bending of a membrane is described within the Euler–Bernoulli theory grounded on two assumptions: static and kinematic. According to the static hypothesis, the normal component T_{11} of the stress tensor $\mathbf{T} = T_{mn} \mathbf{i}_m \otimes \mathbf{i}_n$ ($m, n = 1, 2, 3$) vanishes. Setting

$$\boldsymbol{\epsilon} = \epsilon_{mn} \mathbf{i}_m \otimes \mathbf{i}_n, \quad \bar{\epsilon} = \epsilon_{22} + \epsilon_{33},$$

and using Eqs. (56), (61), we present the condition $T_{11} = 0$ in the form

$$\frac{\nu}{1-2\nu} (\epsilon_{11} + \bar{\epsilon} - p) + \epsilon_{11} + \frac{\varepsilon(1+\nu)}{2E} \left(\frac{\partial \Phi}{\partial X_1} \right)^2 = 0.$$

It follows from this equation that

$$\epsilon_{11} = -\frac{\nu}{1-\nu} (\bar{\epsilon} - p) - \frac{\varepsilon(1+\nu)(1-2\nu)}{2E(1-\nu)} \left(\frac{\partial \Phi}{\partial X_1} \right)^2. \quad (62)$$

Insertion of Eqs. (61), (62) into Eq. (56) implies the formula for the in-plane stress tensor $\bar{\mathbf{T}} = T_{mn} \mathbf{i}_m \otimes \mathbf{i}_n$ ($m, n = 2, 3$)

$$\bar{\mathbf{T}} = \frac{E\nu}{1-\nu^2} (\bar{\epsilon} - p) \bar{\mathbf{I}} + \frac{E}{1+\nu} \bar{\epsilon} - \frac{\varepsilon(2-\nu)}{2(1-\nu)} \left(\frac{\partial \Phi}{\partial X_1} \right)^2 \bar{\mathbf{I}} \quad (63)$$

where $\bar{\epsilon} = \epsilon_{mn} \mathbf{i}_m \otimes \mathbf{i}_n$ ($m, n = 2, 3$) stands for the in-plane strain tensor, and $\bar{\mathbf{I}} = \mathbf{i}_2 \otimes \mathbf{i}_2 + \mathbf{i}_3 \otimes \mathbf{i}_3$ denotes the in-plane unit tensor.

According to the kinematic hypothesis, the components of the displacement vector $\mathbf{u} = u_m \mathbf{i}_m$ are given by

$$u_1 = w, \quad u_2 = \xi X_2 - \frac{\partial w}{\partial X_2} X_1, \quad u_3 = 0,$$

where $\xi(t)$ denotes axial strain, $w(t, X_2)$ stands for deflection. Calculating components ϵ_{mn} of the tensor $\bar{\epsilon}$ by the conventional formulas

$$\epsilon_{mn} = \frac{1}{2} \left(\frac{\partial u_m}{\partial X_n} + \frac{\partial u_n}{\partial X_m} \right),$$

we find that

$$\bar{\epsilon} = (\xi - X_1 \kappa) \mathbf{i}_2 \otimes \mathbf{i}_2 \quad (64)$$

where

$$\kappa = \frac{\partial^2 w}{\partial X_2^2}$$

denotes the curvature of the membrane. It follows from Eqs. (63), (64) that the only nonzero component of the in-plane stress tensor reads

$$T_{22} = \frac{E}{1-\nu^2} (\xi - X_1 \kappa - \nu p) - \frac{\varepsilon(2-\nu)}{2(1-\nu)} \left(\frac{\partial \Phi}{\partial X_1} \right)^2. \quad (65)$$

Introducing tensile force N and bending moment M by the formulas

$$N = \int_{-H}^H T_{22} dX_1, \quad M = \int_{-H}^H T_{22} X_1 dX_1,$$

equating N and M to zero, and using Eq. (65), we obtain

$$\begin{aligned} \xi &= \frac{1}{2H} \int_{-H}^H \left[\nu p + \frac{\varepsilon(2-\nu)(1+\nu)}{2E} \left(\frac{\partial \Phi}{\partial X_1} \right)^2 \right] dX_1, \\ \kappa &= -\frac{3}{2H^3} \int_{-H}^H \left[\nu p + \frac{\varepsilon(2-\nu)(1+\nu)}{2E} \left(\frac{\partial \Phi}{\partial X_1} \right)^2 \right] X_1 dX_1. \end{aligned}$$

Substitution of Eq. (57) into the last equality implies that

$$\kappa = -\frac{3}{2EH^3} \int_{-H}^H \left[\frac{3}{\gamma_0} a k_B T \nu (1-2\nu) (C^+ + C^- - 2C_0) + \frac{\varepsilon}{2} (2-\nu)(1+\nu) \left(\frac{\partial \Phi}{\partial X_1} \right)^2 \right] X_1 dX_1. \quad (66)$$

3.2 Initial and boundary conditions

It follows from Eq. (66) that the curvature of the membrane is determined by the functions C^+ , C^- , Φ that obey Eqs. (58)–(60). The initial conditions for Eqs. (58), (59) read

$$C^+(0, X_1) = C_0, \quad C^-(0, X_1) = C_0. \quad (67)$$

Keeping in mind that the flux of anions through the electrodes is absent, we write

$$J_L^- = 0, \quad J_R^- = 0 \quad (68)$$

where the subscript indices L and R refer to the values of appropriate functions at the left ($X_1 = -H$) and right ($X_1 = H$) ends of the interval $[-H, H]$.

Assuming the electrostatic potential Φ to change linearly across the Stern layers near the electrodes, we adopt the following conditions for the function Φ [60]:

$$\Phi_L - \left(\frac{\partial \Phi}{\partial X_1} \right)_L l_S = -\frac{V}{2}, \quad \Phi_R + \left(\frac{\partial \Phi}{\partial X_1} \right)_R l_S = \frac{V}{2} \quad (69)$$

where l_S denotes the length of the Stern layers.

Unlike anions, cations are presumed to react with the electrodes. The corresponding boundary conditions are given by [60]

$$J_L^+ = R_L, \quad J_R^+ = R_R \quad (70)$$

where R_L is the rate of production of cations at $X_1 = -H$, and R_R is the rate of consumption of cations at $X_1 = H$. The choice of no-flux boundary conditions (68) for anions and boundary conditions accounting for chemical reactions (70) for cations is supported by observations reported in [7, 23, 24].

Following [60, 66], we adopt the simplest model for electrochemical reactions, according to which cations react with electrons to produce neutral atoms immobilized at the electrodes, whereas some metal atoms dissociate into mobile cations and electrons. Under the condition that the rate of consumption of cations is

proportional to their concentration, whereas the rate of their production is constant (an infinite reservoir of atoms), we conclude that

$$R_L = -A_L C_L^+ + B_L, \quad R_R = A_R C_R^+ - B_R \quad (71)$$

where A_L , A_R and B_L , B_R denote rates of oxidation and reduction. With reference to the Butler–Volmer concept [61], these quantities are given by

$$\begin{aligned} A_L &= A \exp\left(-\frac{\alpha e \Delta \Phi_L}{k_B T}\right), \quad B_L = B \exp\left(\frac{\alpha e \Delta \Phi_L}{k_B T}\right), \\ A_R &= A \exp\left(-\frac{\alpha e \Delta \Phi_R}{k_B T}\right), \quad B_R = B \exp\left(\frac{\alpha e \Delta \Phi_R}{k_B T}\right) \end{aligned} \quad (72)$$

where A and B are constants, $\alpha = \frac{1}{2}$ is the transfer coefficient, and potential drops (from inside to outside) across the Stern layers $\Delta \Phi_L$, $\Delta \Phi_R$ are determined by Eq. (69),

$$\Delta \Phi_L = \Phi_L + \frac{V}{2} = \left(\frac{\partial \Phi}{\partial X_1}\right)_L l_S, \quad \Delta \Phi_R = \Phi_R - \frac{V}{2} = -\left(\frac{\partial \Phi}{\partial X_1}\right)_R l_S. \quad (73)$$

Combination of Eqs. (70)–(73) results in

$$\begin{aligned} J_L^+ &= -A C_L^+ \exp\left[-\frac{l_S e}{2k_B T} \left(\frac{\partial \Phi}{\partial X_1}\right)_L\right] + B \exp\left[\frac{l_S e}{2k_B T} \left(\frac{\partial \Phi}{\partial X_1}\right)_L\right], \\ J_R^+ &= A C_R^+ \exp\left[\frac{l_S e}{2k_B T} \left(\frac{\partial \Phi}{\partial X_1}\right)_R\right] - B \exp\left[-\frac{l_S e}{2k_B T} \left(\frac{\partial \Phi}{\partial X_1}\right)_R\right]. \end{aligned} \quad (74)$$

3.3 Steady-state solution

Under conditions (67)–(69) and (74), the curvature of a membrane is determined by five parameters, C_0 , V , A , B , and l_S . To develop an explicit expression for this quantity, we focus on the steady-state solution of Eqs. (58)–(60). Disregarding the terms with the derivatives with respect to time in Eq. (59), we conclude that J^+ and J^- are independent of coordinate X_1 . It follows from Eq. (68) that $J^- = 0$. Under these conditions, Eqs. (58), (60) imply that the functions C^+ , C^- , Φ are governed by the equations

$$\frac{dC_\star^+}{dx} + C_\star^+ \frac{d\Phi_\star}{dx} = -I, \quad \frac{dC_\star^-}{dx} - C_\star^- \frac{d\Phi_\star}{dx} = 0, \quad -\eta^2 \frac{d^2 \Phi_\star}{dx^2} = C_\star^+ - C_\star^- \quad (75)$$

where

$$x = \frac{X_1}{H}, \quad C_\star^+ = \frac{C^+}{C_0}, \quad C_\star^- = \frac{C^-}{C_0}, \quad \Phi_\star = \frac{e\Phi}{k_B T}, \quad I = \frac{J^+ H}{D C_0} \quad (76)$$

denote dimensionless variables,

$$l_D = \sqrt{\frac{\varepsilon k_B T}{C_0 e^2}} \quad (77)$$

stands for the Debye screening length, and

$$\eta = \frac{l_D}{H}. \quad (78)$$

Setting $\varepsilon = \varepsilon_r \varepsilon_0$, where ε_0 is the vacuum permittivity, and ε_r stands for the relative permittivity, we find that for Nafion membranes with ε_r ranging from 20 [70] to 100 [50] containing 1 mol/L of a ionic liquid at room temperature, l_D varies between 0.22 and 0.43 nm. For a membrane with thickness $2H = 0.1$ mm, Eq. (78) implies that $\eta \leq 10^{-3} \ll 1$.

Boundary conditions (69) for the function $\Phi_\star$ read

$$\Phi_\star(-1) - \delta\eta \frac{d\Phi_\star}{dx}(-1) = -\frac{V_\star}{2}, \quad \Phi_\star(1) + \delta\eta \frac{d\Phi_\star}{dx}(1) = \frac{V_\star}{2} \quad (79)$$

where

$$V_\star = \frac{eV}{k_B T}, \quad \delta = \frac{l_S}{l_D}. \quad (80)$$

Parameters $V_\star$ and δ are presumed to be of order of unity. It follows from Eq. (80) that $V_\star = 39.1$ for $V = 1$ V at room temperature. Values of δ are unknown and can vary in a wide interval [60].

Introducing the notation

$$A_\star = \frac{AH}{D}, \quad B_\star = \frac{BH}{DC_0},$$

we find from Eqs. (74) and (79) that

$$\begin{aligned} I &= -A_\star C_\star^+(-1) \exp\left(-\frac{1}{2}\left(\frac{V_\star}{2} + \Phi_\star(-1)\right)\right) + B_\star \exp\left(\frac{1}{2}\left(\frac{V_\star}{2} + \Phi_\star(-1)\right)\right), \\ I &= A_\star C_\star^+(1) \exp\left(\frac{1}{2}\left(\frac{V_\star}{2} - \Phi_\star(1)\right)\right) - B_\star \exp\left(-\frac{1}{2}\left(\frac{V_\star}{2} - \Phi_\star(1)\right)\right). \end{aligned} \quad (81)$$

In dimensionless form, the conservation law for anions reads

$$\int_{-1}^1 C_\star^-(x) dx = 2. \quad (82)$$

Insertion of Eq. (76) into Eq. (66) yields

$$\kappa = \frac{3k_B T C_0}{2EH} \left[\frac{3}{\gamma_0} a\nu(1-2\nu)S_1 + \frac{1}{2}(2-\nu)(1+\nu)S_2 \right] \quad (83)$$

where

$$S_1 = -\int_{-1}^1 (C_\star^+ + C_\star^- - 2)x dx, \quad S_2 = -\eta^2 \int_{-1}^1 \left(\frac{d\Phi_\star}{dx}\right)^2 x dx. \quad (84)$$

Asymptotic analysis of Eq. (75) with conditions (79), (81), (82) is performed in the ‘‘Appendix’’. Combination of Eqs. (83), (84), (A-55), (A-56) implies that the leading term in the expression for the curvature reads

$$\kappa = \frac{3ak_B T C_0}{\gamma_0 E H} \nu(1-2\nu)I \quad (85)$$

where I is a solution of nonlinear equations (A-58), (A-60). It is worth noting that Eq. (85) does not involve the Maxwell stress. To simplify this relation, we apply Eq. (55) to calculate the volume deformation of the membrane swollen with a ionic liquid,

$$\mathcal{E}_0 = \mathcal{E}_0 : \mathbf{I} = \frac{6ak_B T C_0}{\gamma_0 E} (1-2\nu),$$

substitute this expression into Eq. (85), and obtain

$$\kappa = \frac{\mathcal{E}_0 \nu}{2H} I. \quad (86)$$

It follows from Eq. (86) that for a membrane with a fixed thickness H , Poisson’s ratio ν , and volume deformation under swelling $\mathcal{E}_0$, its steady-state curvature κ is determined by the dimensionless parameter I only. To connect this quantity with voltage $V_\star$, we focus on the extreme cases of small and large δ .

3.4 Special cases

When the ratio of thicknesses of the Stern and diffuse boundary layers is small, $\delta \ll 1$, we set $\delta \rightarrow 0$ in Eq. (A-60) and find that

$$\Phi_L = -\frac{V_\star}{2}, \quad \Phi_R = \frac{V_\star}{2}. \quad (87)$$

Substitution of Eq. (87) into Eq. (A-59) yields

$$\exp\left(b + \frac{V_\star}{2}\right) = \frac{2(B_\star - I)}{A_\star(2 + I)^2}, \quad \exp\left(b - \frac{V_\star}{2}\right) = \frac{2(B_\star + I)}{A_\star(2 - I)^2}. \quad (88)$$

It follows from Eqs. (87), (88) that I obeys the nonlinear equation

$$\exp(V_\star) = \frac{B_\star - I}{B_\star + I} \left(\frac{2 - I}{2 + I} \right)^2. \quad (89)$$

For a given voltage $V_\star$, Eq. (89) implies that I is affected by the dimensionless rate $B_\star$ of chemical reactions at electrodes only.

When the ratio δ is sufficiently large, we set $\delta \rightarrow \infty$ in Eq. (A-60) to obtain

$$\exp(b - \Phi_L) = \frac{1}{2 + I}, \quad \exp(b - \Phi_R) = \frac{1}{2 - I}. \quad (90)$$

It follows from Eq. (90) that

$$\exp(\Phi_R - \Phi_L) = \frac{2 - I}{2 + I}. \quad (91)$$

Insertion of Eq. (90) into Eq. (A-59) yields

$$\frac{A_\star(2 + I)}{2} = (B_\star\phi_L - I)\phi_L, \quad \frac{A_\star(2 - I)}{2} = (B_\star\phi_R + I)\phi_R \quad (92)$$

where

$$\phi_L = \exp\left(\frac{1}{2}\left(\frac{V_\star}{2} + \Phi_L\right)\right), \quad \phi_R = \exp\left(-\frac{1}{2}\left(\frac{V_\star}{2} - \Phi_R\right)\right). \quad (93)$$

The solutions of Eq. (92) read

$$\phi_L = \frac{1}{2B_\star} \left(\sqrt{I^2 + 2A_\star B_\star(2 + I)} + I \right), \quad \phi_R = \frac{1}{2B_\star} \left(\sqrt{I^2 + 2A_\star B_\star(2 - I)} - I \right) \quad (94)$$

where

$$R_\star = A_\star B_\star = \frac{AB}{C_0} \left(\frac{H}{D} \right)^2.$$

It follows from Eq. (93) that

$$\left(\frac{\phi_L}{\phi_R} \right)^2 = \exp(V_\star) \exp(-(\Phi_R - \Phi_L)).$$

Substitution of Eqs. (91) and (94) into this equality results in

$$\exp(V_\star) = \frac{2 - I}{2 + I} \left(\frac{\sqrt{I^2 + 2R_\star(2 + I)} + I}{\sqrt{I^2 + 2R_\star(2 - I)} - I} \right)^2. \quad (95)$$

For a fixed voltage $V_\star$, Eq. (95) implies that I is affected by the parameter $R_\star$ only.

4 Numerical simulation

To evaluate the effect of dimensionless voltage V_* on the steady-state curvature of a porous membrane, we solve Eqs. (89) and (95) numerically by the Newton–Raphson algorithm, substitute the results into Eq. (86), and calculate the dimensionless curvature

$$\kappa_* = \frac{2H\kappa}{\mathcal{E}_0\nu} = I. \quad (96)$$

4.1 Thin Stern layer

We begin with analytical estimates (at small and large V_*) for a solution of Eq. (89) that describes curvature of a membrane when the thickness of the Stern layer is substantially lower than that of the diffuse layer, $\delta \ll 1$.

Bearing in mind that $B_* > 0$, it is easy to check that $I = 0$ provides the only solution of Eq. (89) with $V_* = 0$,

$$I|_{V_*=0} = 0.$$

To evaluate I at small voltages V_* , we calculate the derivative dI/dV_* and find that

$$\left. \frac{dI}{dV_*} \right|_{V_*=0} = -\frac{B_*}{2(B_* + 1)}.$$

It follows from these relations and Eq. (96) that κ_* is negative when V_* remains sufficiently small.

When $B_* < 2$, Eq. (89) implies that $V_* \rightarrow \infty$ at $I \rightarrow -B_*$. When $B_* > 2$, it follows from Eq. (89) that $V_* \rightarrow \infty$ at $I \rightarrow -2$. According to Eq. (88), both conditions result in $b \rightarrow \pm\infty$, which means that the solution Φ_0 of Eqs. (A-4.1-3) cannot be presented in the form (A-7). Taking into account Eq. (96), we conclude that at large V_*

$$\kappa_* \rightarrow -B_* \quad (B_* < 2), \quad \kappa_* \rightarrow -2 \quad (B_* > 2).$$

At intermediate values of V_* , Eq. (89) is solved numerically. Results of simulation for various B_* are reported in Fig. 2 where κ_* is plotted versus V_* . The following conclusions are drawn:

1. The steady-state curvature κ_* is negative for all V_* and B_* under investigation (case B in Fig. 1).
2. Given B_* , parameter $|\kappa_*|$ increases monotonically with V_* . The growth of $|\kappa_*|$ is linear under small voltages and becomes sub-linear with an increase in V_* . At relatively large voltages V_* , the steady-state curvature reaches its ultimate value (saturation).
3. Given V_* , parameter $|\kappa_*|$ increases with B_* . This growth is strong at small B_* values ($B_* < 1$) and weakens at large values of this parameter.
4. The critical potential V_* at which an increase in $|\kappa_*|$ levels off grows noticeably with B_* .

To assess how the steady-state curvature of a porous membrane is affected by initial concentration of ions, we fix some characteristic concentration of ionic liquid C_* , set

$$\kappa_* = \frac{\gamma_0 E H \kappa}{3 a k_B T C_* \nu (1 - 2\nu)}, \quad C_{0*} = \frac{C_0}{C_*}, \quad (97)$$

and find from Eq. (85) that

$$\kappa_* = C_{0*} I \quad (98)$$

where I is a solution of Eq. (89) with

$$B_* = \frac{B_*}{C_{0*}}, \quad B_* = \frac{B H}{D C_*}.$$

Results of simulation are depicted in Fig. 3 where κ_* is plotted versus C_{0*} for $B_* = 10.0$ and various V_* . This figure demonstrates that $|\kappa_*|$ increases linearly with C_{0*} for all voltages under consideration, and the rate of growth of $|\kappa_*|$ with C_{0*} increases monotonically with V_* .

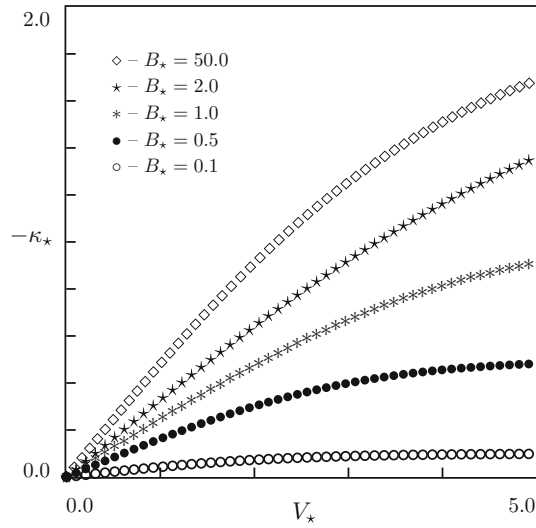


Fig. 2 Curvature κ_* versus voltage V_* . *Symbols*: results of simulation with various B_*

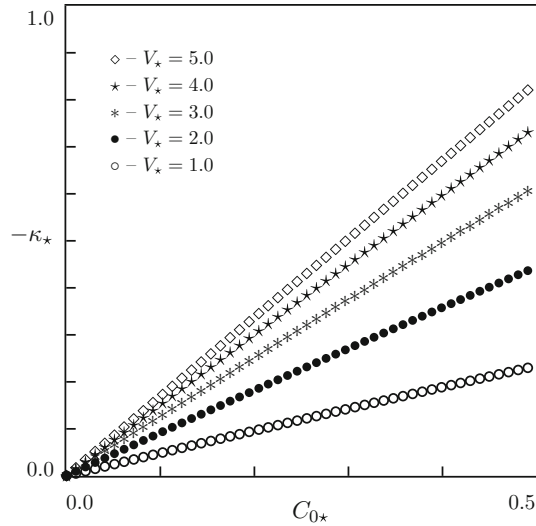


Fig. 3 Curvature κ_* versus concentration of ionic liquid C_{0*} . *Symbols*: results of simulation with $B_* = 10.0$ and various V_*

To examine the effect of thickness on the steady-state curvature of a membrane, we introduce some characteristic thickness H_* , set

$$\kappa_* = \frac{\gamma_0 E H_* \kappa}{3 a k_B T C_0 \nu (1 - 2\nu)}, \quad H_* = \frac{H}{H_*}, \quad (99)$$

and find from Eq. (85) that

$$\kappa_* = \frac{I}{H_*} \quad (100)$$

where I is a solution of Eq. (89) with

$$B_* = B_* H_*, \quad B_* = \frac{B H_*}{D C_0}.$$

Results of a numerical analysis are presented in Fig. 4 where κ_* is plotted versus H_* for $B_* = 1.0$ and various V_* . According to this figure, parameter $|\kappa_*|$ decreases with H_* for all voltages V_* . The decay is pronounced at small V_* , but weakens noticeably with an increase in applied potential.

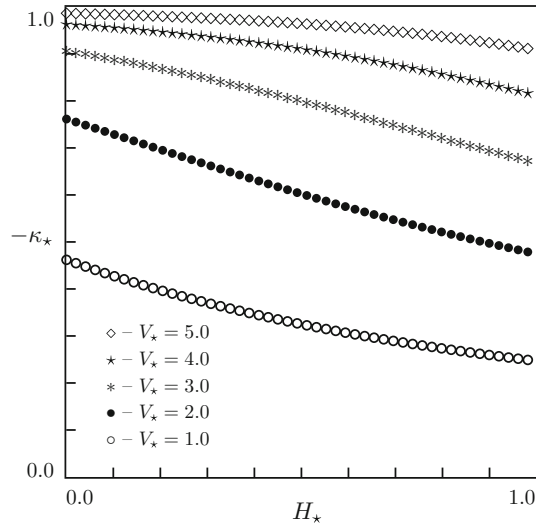


Fig. 4 Curvature κ_* versus thickness H_* . Symbols: results of simulation with $B_* = 1.0$ and various V_*

4.2 Thick Stern layer

To study electroactuation of a porous membrane when thickness of the Stern layer exceeds that of the diffuse layer, $\delta \gg 1$, we begin with analytical estimates for solutions of Eq. (95).

To evaluate I at small voltages V_* , we, first, take into account that $I = 0$ is the only solution of Eq. (95) with $V_* = 0$,

$$I|_{V_*=0} = 0.$$

Calculation of the derivative dI/dV_* implies that

$$\left. \frac{dI}{dV_*} \right|_{V_*=0} = \frac{1}{2} \sqrt{R_*}.$$

These relations together with Eq. (96) imply that κ_* is positive at sufficiently small V_* .

It follows from Eq. (95) that $V_* \rightarrow \infty$ at $I \rightarrow 2$. Bearing in mind Eq. (96), we find that for any $R_* > 0$

$$\kappa_* \rightarrow 2$$

at large voltages V_* .

To describe how the steady-state curvature is affected by voltage at intermediate V_* values, we solve Eq. (95) numerically, calculate κ_* from Eq. (96), and report the results in Figs. 5 and 6, where κ_* is plotted versus dimensionless voltage V_* at large (Fig. 5) and small (Fig. 6) values of R_* . The following conclusions are drawn:

1. At all voltages V_* , the steady-state curvature κ_* is positive (Case A in Fig. 1).
2. This curvature increases monotonically with V_* and levels off at relatively high voltages.
3. At small voltages, κ_* increases linearly with V_* , and the rate of its growth is proportional to $\sqrt{R_*}$.
4. Beyond the interval of linear growth, the function $\kappa_*(V_*)$ increases sub-linearly when R_* exceeds 0.5 and reveals an interval of superlinear growth at $R_* < 0.1$.

To assess the influence of concentration of ionic liquid C_0 on the steady-state curvature of a membrane, we perform a numerical analysis of Eqs. (95), (98) with

$$R_* = \frac{R_*}{C_{0*}}, \quad R_* = \frac{AB}{C_*} \left(\frac{H}{D} \right)^2$$

where C_{0*} is given by Eq. (97). Results of the simulation are reported in Fig. 7 where κ_* is plotted versus C_{0*} for $R_* = 0.1$ and various V_* . This figure shows that the steady-state curvature increases with the initial

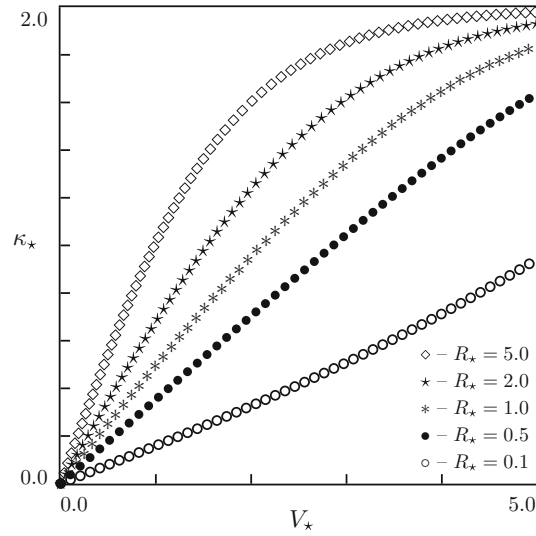


Fig. 5 Curvature κ_* versus voltage V_* . Symbols results of simulation with various R_*

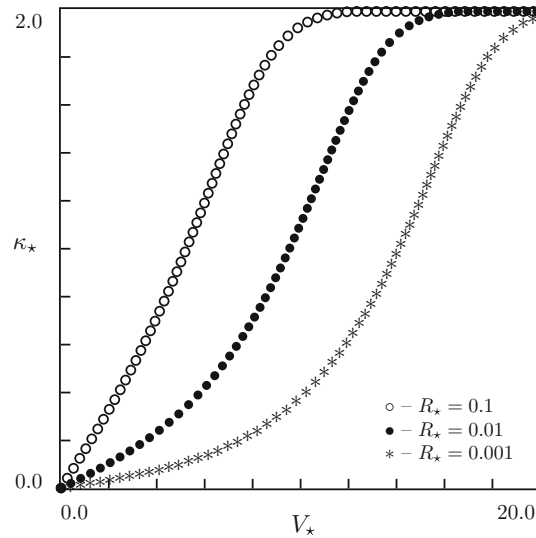


Fig. 6 Curvature κ_* versus voltage V_* . Symbols: results of simulation with various R_*

concentration of ions for all voltages V_* . The growth is practically linear at relatively small V_* and becomes sub-linear when V_* exceeds 3.0.

To examine the effect of thickness on the curvature of a membrane, we solve Eqs. (95), (100) with

$$R_* = R_* H_*^2, \quad R_* = \frac{AB}{C_0} \left(\frac{H_*}{D} \right)^2$$

where H_* stands for the characteristic thickness, and H_* is given by Eq. (99). The simulation is performed with $R_* = 0.1$ and various V_* . Parameter κ_* is plotted versus H_* in Fig. 8. According to this figure, the steady-state curvature of a porous membrane grows linearly with its thickness for all voltages. The rate of growth of the function $\kappa_*(H_*)$ increases with voltage V_* .

4.3 Comparison with observations

In accord with the experimental data discussed in the Introduction, simulation reveals that the steady-state curvature of a porous membrane can be positive, as well as negative. According to the model, curvature κ_s is

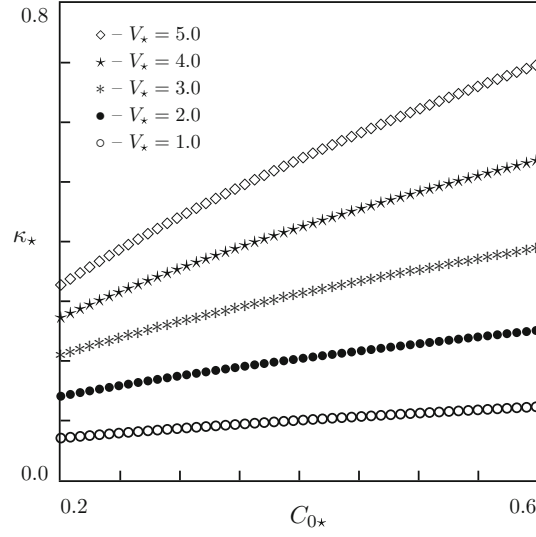


Fig. 7 Curvature κ_* versus concentration of ionic liquid C_{0*} . Symbols: results of simulation with $R_* = 0.1$ and various V_*

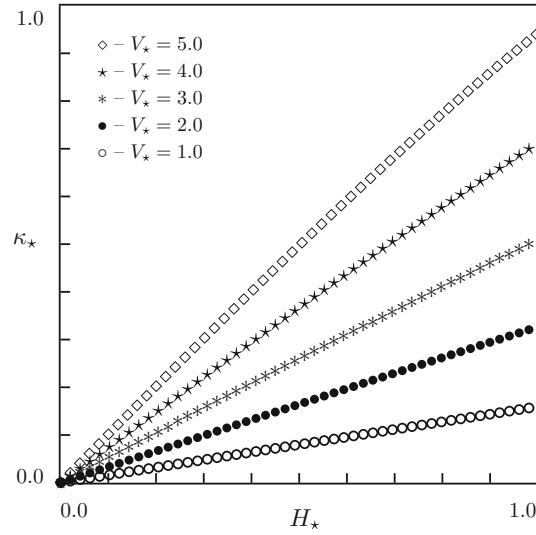


Fig. 8 Curvature κ_* versus thickness H_* . Symbols: results of simulation with $R_* = 0.1$ and various V_*

positive when the Stern layer is sufficiently thick, and it is negative when the thickness of this layer becomes negligible compared with that of the diffuse layer.

The model predicts that the absolute value of steady-state curvature increases linearly with applied voltage when V_* remains sufficiently small and levels off at some ultimate values when V_* becomes large. In between these extremes, the dependence $\kappa_s(V_*)$ is sub-linear for thin Stern layers, and it becomes S-shaped for thick Stern layers and $R_* < 0.5$. All conclusions are in qualitative agreement with observations.

Results of the numerical analysis demonstrate that the absolute value of steady-state curvature grows with concentration of ionic liquid. When κ_* is negative, the increase in curvature with C_0 is linear, and its rate is proportional to applied voltage. For positive κ_* , this increase is linear at small voltages, and it is transformed into a sub-linear growth when V_* increases. The growth of curvature with C_0 is in accord with experimental data reported in [25,29]. As our analysis is confined to small strains under swelling and electroactuation, simulation does not reveal a decay in κ_* with C_0 at high concentrations of ionic liquid [31,32]. Formally, this reduction may be taken into account by presuming porosity γ_0 in Eq. (85) to grow with C_0 . However, this approach should be treated with caution as in highly swollen membranes other parameters in the governing equations depend on C_0 as well.

Simulation reveals that the absolute value of steady-state curvature decreases with the thickness of a porous membrane when the Stern layer is thin. For membranes with relatively large δ , the steady-state curvature increases linearly with H_* , and the rate of its growth is strongly enhanced by voltage. The latter conclusion is in accord with observations reported in [33,34].

5 Conclusions

A model is developed for electroactuation of a porous elastic membrane swollen with a ionic liquid and sandwiched between nanoscale-thin electrodes (whose bending rigidity is disregarded). Bending of the membrane under applied voltage is attributed to ionic pressure induced by diffusion of ions through the membrane. Transport of ions is driven by the electric field and redox reactions at the surfaces of the electrodes.

Constitutive equations for the elastic response of a porous medium and diffusion of ions are derived by means of the free energy imbalance inequality for an arbitrary three-dimensional deformation with finite strains. After simplification for the case of small strains, but finite changes in the electrostatic potential and concentrations of ions, these relations are applied to describe plane bending of a membrane. Curvature of a porous membrane is expressed in terms of a solution of one-dimensional Poisson–Nernst–Planck equations for concentrations of ions and electrostatic potential. The steady-state solution of these equations is obtained by means of the method of matched asymptotic expansion.

The main result of this study is Eq. (85) that expresses the steady-state curvature of a membrane in terms of a solution of four nonlinear equations (A-59), (A-60). For the limiting cases of small and large thicknesses of the Stern layer (a compact layer in the close vicinity of an electrode where chemical reactions occur), the set of nonlinear equations is reduced to one equation only, which is solved numerically by using the Newton–Raphson algorithm.

Simulation reveals that the steady-state curvature of a porous membrane is positive when the Stern layer is sufficiently thick and negative when its thickness becomes negligible. The absolute value of this curvature increases with applied electrostatic potential at small voltages and levels off at large voltages. An increase in concentration of ionic liquid results in the growth of curvature, whereas an increase in thickness of the membrane enhances bending to the anode and weakens bending to the cathode. All conclusions are in qualitative agreement with available experimental data.

To develop a tractable model for electroactuation of a porous membrane and to obtain its analytical solution, several phenomena have been neglected, and a number of simplifications have been made. In particular, we disregard bound charges attached to polymer chains, neglect interactions between finite-sized ions, presume a ionic liquid to dissociate entirely into positive and negative ions, apply an overly simplified model for evolution of porosity, confine ourselves to the study of small strains under swelling in an ionic liquid and to the steady-state regime of bending, and neglect the effect of concentration of ions on their diffusivities and electric permittivity of the membrane.

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Appendix

To simplify the notation, we omit stars as subscripts of the unknown functions and present Eq. (75) in the form

$$\frac{dC^+}{dx} + C^+ \frac{d\Phi}{dx} = -I, \quad \frac{dC^-}{dx} - C^- \frac{d\Phi}{dx} = 0, \quad -\eta^2 \frac{d^2\Phi}{dx^2} = C^+ - C^-. \quad (\text{A-1})$$

The boundary conditions for Eq. (A-1) are given by

$$C^+(-1) = C_L^+, \quad C^+(1) = C_R^+, \quad \Phi(-1) = \Phi_L, \quad \Phi(1) = \Phi_R \quad (\text{A-2})$$

where Φ_L , Φ_R and C_L^+ , C_R^+ obey Eqs. (79) and (81).

Following [67], we begin with the derivation of an approximate solution in the interior of the interval $(-1, 1)$, where the functions C^+ , C^- , Φ are expanded into regular series in a small parameter η ,

$$C^+ = \bar{C}_0^+ + \eta \bar{C}_1^+ + \dots, \quad C^- = \bar{C}_0^- + \eta \bar{C}_1^- + \dots, \quad \Phi = \bar{\Phi}_0 + \eta \bar{\Phi}_1 + \dots. \quad (\text{A-3})$$

Substitution of Eq. (A-3) into Eq. (A-1) implies that

$$\frac{d\bar{C}_0^+}{dx} + \bar{C}_0^+ \frac{d\bar{\Phi}_0}{dx} = -I, \quad \frac{d\bar{C}_0^-}{dx} - \bar{C}_0^- \frac{d\bar{\Phi}_0}{dx} = 0, \quad 0 = \bar{C}_0^+ - \bar{C}_0^-. \quad (\text{A-4.1-3})$$

It follows from Eq. (A-4.3) that

$$\bar{C}_0^- = \bar{C}_0^+. \quad (\text{A-5})$$

Inserting expression (A-5) into Eq. (A-4.1, 2) and resolving them with respect to the derivatives, we find that

$$\frac{d\bar{C}_0^+}{dx} = -\frac{I}{2}, \quad \frac{d\bar{\Phi}_0}{dx} = -\frac{I}{2\bar{C}_0^+}.$$

Integration of these equations implies that

$$\bar{C}_0^+ = \frac{a - Ix}{2} \quad (\text{A-6})$$

and

$$\bar{\Phi}_0 = \ln(a - Ix) + b \quad (\text{A-7})$$

where a and b are unknown constants. Equations (A-5)–(A-7) express the leading terms in the regular part of the solution by means of three parameters: I , a , b .

We now consider the boundary layer near the point $x = -1$ and search for a singular part of the solution in the form

$$C^+ = C_l^+(\zeta), \quad C^- = C_l^-(\zeta), \quad \Phi = \Phi_l(\zeta) \quad (\text{A-8})$$

where

$$\zeta = \frac{1 + x}{\eta} \quad (\text{A-9})$$

stands for the stretched coordinate. Expanding the functions C_l^+ , C_l^- , Φ_l into series in η ,

$$C_l^+ = C_{l0}^+ + \eta c_{l1}^+ + \dots, \quad C_l^- = C_{l0}^- + \eta c_{l1}^- + \dots, \quad \Phi_l = \Phi_{l0} + \eta \Phi_{l1} + \dots, \quad (\text{A-10})$$

inserting Eq. (A-10) into Eq. (A-1), and keeping in mind that

$$\frac{d}{dx} = \frac{1}{\eta} \frac{d}{d\zeta},$$

we conclude that the leading terms in series (A-10) obey the equations

$$\frac{dC_{l0}^+}{d\zeta} + C_{l0}^+ \frac{d\Phi_{l0}}{d\zeta} = 0, \quad \frac{dC_{l0}^-}{d\zeta} - C_{l0}^- \frac{d\Phi_{l0}}{d\zeta} = 0, \quad \frac{d^2\Phi_{l0}}{d\zeta^2} = C_{l0}^- - C_{l0}^+ \quad (\text{A-11.1-3})$$

with boundary conditions (A-2),

$$C_{l0}^+(0) = C_L^+, \quad C_{l0}^-(0) = C_L^-, \quad \Phi_{l0}(0) = \Phi_L \quad (\text{A-12})$$

where C_L^- is an unknown parameter. It follows from Eq. (A-11.1) that

$$\frac{dC_{l0}^+}{C_{l0}^+} = -d\Phi_{l0}.$$

Integrating this equality from $\zeta = 0$ to ζ and using Eq. (A-12), we find that

$$C_{l0}^+ = C_L^+ \exp\left(-(\Phi_{l0} - \Phi_L)\right). \quad (\text{A-13})$$

Similarly, integration of the other equation in Eq. (A-11.1-3) with boundary condition (A-12) yields

$$C_{l0}^- = C_L^- \exp(\Phi_{l0} - \Phi_L). \quad (\text{A-14})$$

Substitution of Eqs. (A-5), (A-6), (A-7), (A-13), (A-14) into the matching conditions

$$\lim_{\zeta \rightarrow \infty} C_{l0}^+ = \lim_{x \rightarrow -1} \bar{C}_0^+, \quad \lim_{\zeta \rightarrow \infty} C_{l0}^- = \lim_{x \rightarrow -1} \bar{C}_0^-, \quad \lim_{\zeta \rightarrow \infty} \Phi_{l0} = \lim_{x \rightarrow -1} \bar{\Phi}_0 \quad (\text{A-15})$$

implies that

$$C_L^+ = \frac{(a+I)^2}{2} \exp(b - \Phi_L), \quad C_L^- = \frac{1}{2} \exp(-(b - \Phi_L)). \quad (\text{A-16})$$

Insertion of Eqs. (A-13), (A-14), (A-16) into Eq. (A-11.3) results in the differential equation

$$\frac{d^2 \Phi_{l0}}{d\zeta^2} = \frac{1}{2} \left[\exp(\Phi_{l0} - b) - (a+I)^2 \exp(-(\Phi_{l0} - b)) \right]. \quad (\text{A-17})$$

We multiply Eq. (A-17) by $d\Phi_{l0}/d\zeta$, integrate the result from $\zeta = 0$ to ζ , use Eq. (A-12), and obtain

$$\begin{aligned} \left(\frac{d\Phi_{l0}}{d\zeta} \right)^2 &= \left(\frac{d\Phi_{l0}}{d\zeta} \right)_{\zeta=0}^2 + \left[\exp(\Phi_{l0} - b) - \exp(\Phi_L - b) \right] \\ &\quad + (a+I)^2 \left[\exp(-(\Phi_{l0} - b)) - \exp(-(\Phi_L - b)) \right]. \end{aligned} \quad (\text{A-18})$$

It follows from Eqs. (A-15), (A-18) and the matching condition

$$\lim_{\zeta \rightarrow \infty} \frac{d\Phi_{l0}}{d\zeta} = 0 \quad (\text{A-19})$$

that

$$\left(\frac{d\Phi_{l0}}{d\zeta} \right)_{\zeta=0}^2 = \left[\exp\left(\frac{1}{2}(\Phi_L - b)\right) - (a+I) \exp\left(-\frac{1}{2}(\Phi_L - b)\right) \right]^2. \quad (\text{A-20})$$

Substitution of Eq. (A-20) into Eq. (A-18) yields

$$\frac{d\Phi_{l0}}{d\zeta} = \pm \left[\exp\left(\frac{1}{2}(\Phi_{l0} - b)\right) - (a+I) \exp\left(-\frac{1}{2}(\Phi_{l0} - b)\right) \right]. \quad (\text{A-21})$$

Choosing the negative sign in the right-hand side of Eq. (A-21), we obtain

$$\frac{d\Phi_{l0}}{d\zeta} = - \left[\exp\left(\frac{1}{2}(\Phi_{l0} - b)\right) - (a+I) \exp\left(-\frac{1}{2}(\Phi_{l0} - b)\right) \right], \quad (\text{A-22})$$

which implies that

$$\left(\frac{d\Phi_{l0}}{d\zeta} \right)_{\zeta=0} = - \left[\exp\left(\frac{1}{2}(\Phi_L - b)\right) - (a+I) \exp\left(-\frac{1}{2}(\Phi_L - b)\right) \right]. \quad (\text{A-23})$$

Introducing the function

$$Z = \exp\left(\frac{1}{2}(\Phi_{l0} - b)\right), \quad (\text{A-24})$$

we present Eq. (A-22) in the form

$$\frac{dZ}{Z^2 - Z_+^2} = -\frac{1}{2} d\zeta \quad (\text{A-25})$$

where

$$Z_+ = \sqrt{a+I}. \quad (\text{A-26})$$

Integration of Eq. (A-25) with the boundary condition $Z(0) = Z_0$ implies that

$$Z = Z_+ \frac{1 + \Lambda_L \exp(-Z_+ \zeta)}{1 - \Lambda_L \exp(-Z_+ \zeta)} \quad (\text{A-27})$$

where

$$\Lambda_L = \frac{Z_0 - Z_+}{Z_0 + Z_+}. \quad (\text{A-28})$$

According to Eqs. (A-12), (A-24),

$$Z_0 = \exp\left(\frac{1}{2}(\Phi_L - b)\right).$$

Insertion of this expression and Eq. (A-26) into Eq. (A-28) implies that

$$\Lambda_L = \frac{\exp(\frac{1}{2}(\Phi_L - b)) - \sqrt{a + I}}{\exp(\frac{1}{2}(\Phi_L - b)) + \sqrt{a + I}}. \quad (\text{A-29})$$

It follows from Eqs. (A-22), (A-24) that

$$\frac{d\Phi_{I0}}{d\zeta} = -\frac{Z^2 - Z_+^2}{Z}.$$

Combination of this relation with Eq. (A-27) yields

$$\frac{d\Phi_{I0}}{d\zeta} = -\frac{4\Lambda_L Z_+ \exp(-Z_+ \zeta)}{1 - \Lambda_L^2 \exp(-2Z_+ \zeta)}. \quad (\text{A-30})$$

For $\Lambda_L > 0$, which is equivalent to the condition

$$\ln(a + I) + b < \Phi_L,$$

Eq. (A-30) implies that Φ_{I0} decreases monotonically with ζ from Φ_L down to $\ln(a + I) + b$. When $\Lambda_L < 0$, which means that

$$\ln(a + I) + b > \Phi_L,$$

this relation ensures that Φ_{I0} increases with ζ from Φ_L up to $\ln(a + I) + b$.

It follows from Eq. (A-30) that

$$\int_0^\infty \left(\frac{d\Phi_{I0}}{d\zeta}\right)^2 d\zeta = 16\Lambda_L^2 Z_+^2 \int_0^\infty \frac{\exp(-2Z_+ \zeta)}{(1 - \Lambda_L^2 \exp(-2Z_+ \zeta))^2} d\zeta.$$

Setting

$$\zeta_1 = \exp(-2Z_+ \zeta),$$

we find that

$$\int_0^\infty \left(\frac{d\Phi_{I0}}{d\zeta}\right)^2 d\zeta = 8\Lambda_L^2 Z_+ \int_0^1 \frac{d\zeta_1}{(1 - \Lambda_L^2 \zeta_1)^2} = \frac{8\Lambda_L^2 Z_+}{1 - \Lambda_L^2}. \quad (\text{A-31})$$

Combination of Eqs. (A-13), (A-14), (A-16), (A-24) results in

$$C_{I0}^+ = \frac{(a + I)^2}{2Z^2}, \quad C_{I0}^- = \frac{Z^2}{2}. \quad (\text{A-32})$$

It follows from Eqs. (A-27), (A-32) that concentrations C_{I0}^+ , C_{I0}^- evolve monotonically with ζ from their boundary values at $\zeta = 0$ to their values at $\zeta = \infty$ given by Eq. (A-15).

We now turn to the analysis of the right boundary layer near the point $x = 1$ and search for a singular part of the solution in the form

$$C^+ = C_r^+(\zeta), \quad C^- = C_r^-(\zeta), \quad \Phi = \Phi_r(\zeta) \quad (\text{A-33})$$

where

$$\zeta = \frac{1-x}{\eta} \quad (\text{A-34})$$

denotes the stretched coordinate. Expanding the functions C_r^+ , C_r^- , Φ_r into series in η ,

$$C_r^+ = C_{r0}^+ + \eta C_{r1}^+ + \dots, \quad C_r^- = C_{r0}^- + \eta C_{r1}^- + \dots, \quad \Phi_r = \Phi_{r0} + \eta \Phi_{r1} + \dots, \quad (\text{A-35})$$

substituting Eq. (A-35) into Eq. (A-1), and keeping in mind that

$$\frac{d}{dx} = -\frac{1}{\eta} \frac{d}{d\zeta},$$

we find that the leading terms in series (A-35) obey Eq. (A-11),

$$\frac{dC_{r0}^+}{d\zeta} + C_{r0}^+ \frac{d\Phi_{r0}}{d\zeta} = 0, \quad \frac{dC_{r0}^-}{d\zeta} - C_{r0}^- \frac{d\Phi_{r0}}{d\zeta} = 0, \quad \frac{d^2\Phi_{r0}}{d\zeta^2} = C_{r0}^- - C_{r0}^+, \quad (\text{A-36.1-3})$$

with boundary conditions (A-2)

$$C_{r0}^+(0) = C_R^+, \quad C_{r0}^-(0) = C_R^-, \quad \Phi_{r0}(0) = \Phi_R \quad (\text{A-37})$$

where C_R^- is an unknown parameter. Integration of Eq. (A-36.1, 2) with conditions (A-37) results in

$$C_{r0}^+ = C_R^+ \exp\left(-(\Phi_{r0} - \Phi_R)\right), \quad C_{r0}^- = C_R^- \exp(\Phi_{r0} - \Phi_R). \quad (\text{A-38})$$

Insertion of Eqs. (A-5), (A-6), (A-7), (A-38) into the matching conditions

$$\lim_{\zeta \rightarrow \infty} C_{r0}^+ = \lim_{x \rightarrow 1} \bar{C}_0^+, \quad \lim_{\zeta \rightarrow \infty} C_{r0}^- = \lim_{x \rightarrow 1} \bar{C}_0^-, \quad \lim_{\zeta \rightarrow \infty} \Phi_{r0} = \lim_{x \rightarrow 1} \bar{\Phi}_0 \quad (\text{A-39})$$

implies that

$$C_R^+ = \frac{(a-I)^2}{2} \exp(b - \Phi_R), \quad C_R^- = \frac{1}{2} \exp\left(-(b - \Phi_R)\right). \quad (\text{A-40})$$

Substitution of Eqs. (A-38), (A-40) into Eq. (A-36.3) yields

$$\frac{d^2\Phi_{r0}}{d\zeta^2} = \frac{1}{2} \left[\exp(\Phi_{r0} - b) - (a-I)^2 \exp\left(-(\Phi_{r0} - b)\right) \right]. \quad (\text{A-41})$$

Multiplying Eq. (A-41) by $d\Phi_{r0}/d\zeta$, integrating the result from $\zeta = 0$ to ζ , using Eq. (A-37), and calculating the integral, we find that

$$\begin{aligned} \left(\frac{d\Phi_{r0}}{d\zeta} \right)^2 &= \left(\frac{d\Phi_{r0}}{d\zeta} \right)_{\zeta=0}^2 + \left[\exp(\Phi_{r0} - b) - \exp(\Phi_R - b) \right] \\ &\quad + (a-I)^2 \left[\exp\left(-(\Phi_{r0} - b)\right) - \exp\left(-(\Phi_R - b)\right) \right]. \end{aligned} \quad (\text{A-42})$$

It follows from Eqs. (A-39), (A-42) and the matching condition

$$\lim_{\zeta \rightarrow \infty} \frac{d\Phi_{r0}}{d\zeta} = 0 \quad (\text{A-43})$$

that

$$\left(\frac{d\Phi_{r0}}{d\zeta} \right)_{\zeta=0}^2 = \left[\exp\left(\frac{1}{2}(\Phi_R - b)\right) - (a-I) \exp\left(-\frac{1}{2}(\Phi_R - b)\right) \right]^2. \quad (\text{A-44})$$

Substitution of Eq. (A-44) into Eq. (A-42) results in

$$\frac{d\Phi_{r0}}{d\zeta} = \pm \left[\exp\left(\frac{1}{2}(\Phi_{r0} - b)\right) - (a - I) \exp\left(-\frac{1}{2}(\Phi_{r0} - b)\right) \right].$$

We choose the negative sign in the right-hand side of this equation,

$$\frac{d\Phi_{r0}}{d\zeta} = - \left[\exp\left(\frac{1}{2}(\Phi_{r0} - b)\right) - (a - I) \exp\left(-\frac{1}{2}(\Phi_{r0} - b)\right) \right], \quad (\text{A-45})$$

which implies that

$$\left(\frac{d\Phi_{r0}}{d\zeta}\right)_{\zeta=0} = - \left[\exp\left(\frac{1}{2}(\Phi_R - b)\right) - (a - I) \exp\left(-\frac{1}{2}(\Phi_R - b)\right) \right]. \quad (\text{A-46})$$

By analogy with Eq. (A-24), introducing the function

$$Z = \exp\left(\frac{1}{2}(\Phi_{r0} - b)\right), \quad (\text{A-47})$$

we present the solution of Eq. (A-45) in the form (A-27),

$$Z = Z_- \frac{1 + \Lambda_R \exp(-Z_- \zeta)}{1 - \Lambda_R \exp(-Z_- \zeta)}, \quad (\text{A-48})$$

where

$$Z_- = \sqrt{a - I}, \quad \Lambda_R = \frac{\exp(\frac{1}{2}(\Phi_R - b)) - \sqrt{a - I}}{\exp(\frac{1}{2}(\Phi_R - b)) + \sqrt{a - I}}. \quad (\text{A-49})$$

Combination of Eqs. (A-45)–(A-48) yields

$$\frac{d\Phi_{r0}}{d\zeta} = - \frac{4\Lambda_R Z_- \exp(-Z_- \zeta)}{1 - \Lambda_R^2 \exp(-2Z_- \zeta)}. \quad (\text{A-50})$$

For $\Lambda_R > 0$, which is equivalent to the condition

$$\ln(a - I) + b < \Phi_R,$$

Eq. (A-50) implies that Φ_{r0} decreases monotonically with ζ from Φ_R down to $\ln(a - I) + b$. When $\Lambda_R < 0$, which means that

$$\ln(a - I) + b > \Phi_R,$$

Eq. (A-50) demonstrates that Φ_{r0} increases with ζ from Φ_R up to $\ln(a - I) + b$.

It follows from Eq. (A-50) that

$$\int_0^\infty \left(\frac{d\Phi_{r0}}{d\zeta}\right)^2 d\zeta = 16\Lambda_R^2 Z_-^2 \int_0^\infty \frac{\exp(-2Z_- \zeta)}{(1 - \Lambda_R^2 \exp(-2Z_- \zeta))^2} d\zeta.$$

Calculating the integral by analogy with Eq. (A-31), we obtain

$$\int_0^\infty \left(\frac{d\Phi_{r0}}{d\zeta}\right)^2 d\zeta = \frac{8\Lambda_R^2 Z_-}{1 - \Lambda_R^2}. \quad (\text{A-51})$$

It follows from Eqs. (A-38), (A-40), (A-47) that

$$C_{r0}^+ = \frac{(a - I)^2}{2Z^2}, \quad C_{r0}^- = \frac{Z^2}{2}. \quad (\text{A-52})$$

Equations (A-48), (A-52) imply that concentrations C_{r0}^+ , C_{r0}^- evolve monotonically with ζ from their boundary values at $\zeta = 0$ to their values at $\zeta = \infty$ determined by Eq. (A-39).

Our aim now is to calculate the integrals in Eqs. (82) and (84). We begin with Eq. (82) and split the interval of integration $[-1, 1]$ into three parts,

$$\int_{-1}^1 C^- dx = \int_{-1}^{-1+\Delta} C^- dx + \int_{-1+\Delta}^{1-\Delta} C^- dx + \int_{1-\Delta}^1 C^- dx, \quad (\text{A-53})$$

where Δ is a small positive parameter. Keeping in mind that C^- is a continuous bounded function, we conclude that the first and the last integrals in Eq. (A-53) tend to zero with Δ , whereas C^- in the second integral can be replaced with $\bar{C}_0^-$. Disregarding small terms, we present the conservation law (82) in the form

$$\int_{-1}^1 \bar{C}_0^- dx = 2. \quad (\text{A-54})$$

Inserting Eqs. (A-5), (A-6) into Eq. (A-54) and performing integration, we find that

$$a = 2. \quad (\text{A-55})$$

To find S_1 , we split the first integral in Eq. (84) by analogy with Eq. (A-53),

$$S_1 = - \int_{-1}^{-1+\Delta} (C^+ + C^- - 2)x dx - \int_{-1+\Delta}^{1-\Delta} (C^+ + C^- - 2)x dx - \int_{1-\Delta}^1 (C^+ + C^- - 2)x dx,$$

disregard small terms corresponding to the boundary layers, and replace C^+ , C^- in the second integral with the leading terms in expansion (A-3). Taking the limit as $\Delta, \eta \rightarrow 0$, we find that

$$S_1 = - \int_{-1}^1 (\bar{C}_0^+ + \bar{C}_0^- - 2)x dx.$$

Insertion of Eqs. (A-5), (A-6), (A-55) into this formula implies that

$$S_1 = \int_{-1}^1 I x^2 dx = \frac{2}{3} I. \quad (\text{A-56})$$

To determine S_2 , a similar approach is applied. By splitting the last integral in Eq. (84), we find that

$$S_2 = -\eta^2 \int_{-1}^{-1+\Delta} \left(\frac{d\Phi_\star}{dx} \right)^2 x dx - \eta^2 \int_{-1+\Delta}^{1-\Delta} \left(\frac{d\Phi_\star}{dx} \right)^2 x dx - \eta^2 \int_{1-\Delta}^1 \left(\frac{d\Phi_\star}{dx} \right)^2 x dx.$$

The second integral is negligible small as the leading term in the expression for $d\Phi_\star/dx$ is a continuous function, see Eq. (A-7). By the same reasoning, we disregard Φ_0 in the expression for $\Phi_\star$ in the first and last integrals to obtain

$$S_2 = -\eta^2 \int_{-1}^{-1+\Delta} \left(\frac{d\Phi_{l0}}{dx} \right)^2 x dx - \eta^2 \int_{1-\Delta}^1 \left(\frac{d\Phi_{r0}}{dx} \right)^2 x dx.$$

Introducing stretched variables by means of Eqs. (A-9), (A-34), we find that

$$S_2 = -\eta \int_0^{\frac{\Delta}{\eta}} \left[\left(\frac{d\Phi_{l0}}{d\zeta} \right)^2 (-1 + \eta\zeta) + \left(\frac{d\Phi_{r0}}{d\zeta} \right)^2 (1 - \eta\zeta) \right] d\zeta.$$

Disregarding terms beyond the first order of smallness with respect to η and taking the limit as $\Delta/\eta \rightarrow \infty$, we arrive at the formula

$$S_2 = \eta \int_0^\infty \left[\left(\frac{d\Phi_{l0}}{d\zeta} \right)^2 - \left(\frac{d\Phi_{r0}}{d\zeta} \right)^2 \right] d\zeta \quad (\text{A-57})$$

where the integrals are given by Eqs. (A-31), (A-51). It follows from Eq. (A-57) that S_2 is small compared with unity, and it may be disregarded in Eq. (83).

According to Eqs. (83) and (A-56), the curvature κ is uniquely determined by parameter I . Our aim now is to find this quantity from boundary conditions (79), (81). Substituting Eqs. (A-16), (A-40), (A-55) into Eq. (81) and keeping in mind that

$$C_{\star}^{+}(-1) = C_L, \quad C_{\star}^{+}(1) = C_R, \quad \Phi_{\star}(-1) = \Phi_L, \quad \Phi_{\star}(1) = \Phi_R,$$

we obtain

$$\begin{aligned} I &= -\frac{1}{2}A_{\star}(2+I)^2 \exp(b - \Phi_L) \exp\left(-\frac{1}{2}\left(\frac{V_{\star}}{2} + \Phi_L\right)\right) + B_{\star} \exp\left(\frac{1}{2}\left(\frac{V_{\star}}{2} + \Phi_L\right)\right), \\ I &= \frac{1}{2}A_{\star}(2-I)^2 \exp(b - \Phi_R) \exp\left(\frac{1}{2}\left(\frac{V_{\star}}{2} - \Phi_R\right)\right) - B_{\star} \exp\left(-\frac{1}{2}\left(\frac{V_{\star}}{2} - \Phi_R\right)\right). \end{aligned} \quad (\text{A-58})$$

It follows from Eq. (A-58) that

$$\begin{aligned} \exp(b - \Phi_L) &= \frac{2}{A_{\star}(2+I)^2} \left[B_{\star} \exp\left(\frac{1}{2}\left(\frac{V_{\star}}{2} + \Phi_L\right)\right) - I \right] \exp\left(\frac{1}{2}\left(\frac{V_{\star}}{2} + \Phi_L\right)\right), \\ \exp(b - \Phi_R) &= \frac{2}{A_{\star}(2-I)^2} \left[B_{\star} \exp\left(-\frac{1}{2}\left(\frac{V_{\star}}{2} - \Phi_R\right)\right) + I \right] \exp\left(-\frac{1}{2}\left(\frac{V_{\star}}{2} - \Phi_R\right)\right). \end{aligned} \quad (\text{A-59})$$

Bearing in mind that

$$\eta \frac{d\Phi_{\star}}{dx}(-1) = \left(\frac{d\Phi_{l0}}{d\zeta} \right)_{\zeta=0}, \quad \eta \frac{d\Phi_{\star}}{dx}(1) = -\left(\frac{d\Phi_{r0}}{d\zeta} \right)_{\zeta=0},$$

and inserting Eqs. (A-23), (A-46), (A-55) into Eq. (79), we find that

$$\begin{aligned} \frac{V_{\star}}{2} + \Phi_L &= \delta \left[(2+I) \exp(b - \Phi_L) - 1 \right] \exp\left(-\frac{1}{2}(b - \Phi_L)\right), \\ \frac{V_{\star}}{2} - \Phi_R &= \delta \left[1 - (2-I) \exp(b - \Phi_R) \right] \exp\left(-\frac{1}{2}(b - \Phi_R)\right). \end{aligned} \quad (\text{A-60})$$

Equations (A-59), (A-60) provide a set of four nonlinear equations for four unknown I , b , Φ_L , Φ_R to be solved numerically for an arbitrary δ .

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